Master's Comprehensive Examination

Part II | Applied (90 minutes) | December 6, 2003

Instructions: Answer all four questions. Open book and open notes.

1. In one stage of the development of a new drug for an allergy, an experiment is conducted to study how different dosages of the drug affect the duration of relief from allergic symptoms. Ten patients are included in the experiment. Each patient receives a specified dosage of the drug and is asked to report back as soon as the protection of the drug seems to wear off. The observations are reported in the table below which shows the Dosage (X) in milligrams and the Duration of Relief (Y) in days for the ten patients.

Dosage X	Duration of Relief Y
3	9
3	5
4	12
5	9
6	14
6	16
7	22
8	18
8	24
9	22

(a) Plot the data and describe how you would assess the appropriateness of a linear regression model for these data.

(b) A regression model was fitted giving $\hat{a} = -1.07$ and $\hat{b} = 2.74$. In other words the least squares regression line is given by $\hat{y} = -1.07 + 2.74x$. Add this line to your plot. Visually examine the residuals and comment about their distribution.

(c) Do the data constitute strong evidence that the duration of relief increases with higher dosages of the drug? State clearly any assumptions you are making. (Note that the estimated variance of $\hat{b}$ is 0.1945)

(d) Finally, obtain and interpret a 95% confidence interval for the average duration of relief at a dosage of 6 milligrams.